Chapter 14 Review

Campbell Biology, AP Edition, 12e

Chapter 14 Review

Summary of Key Concepts

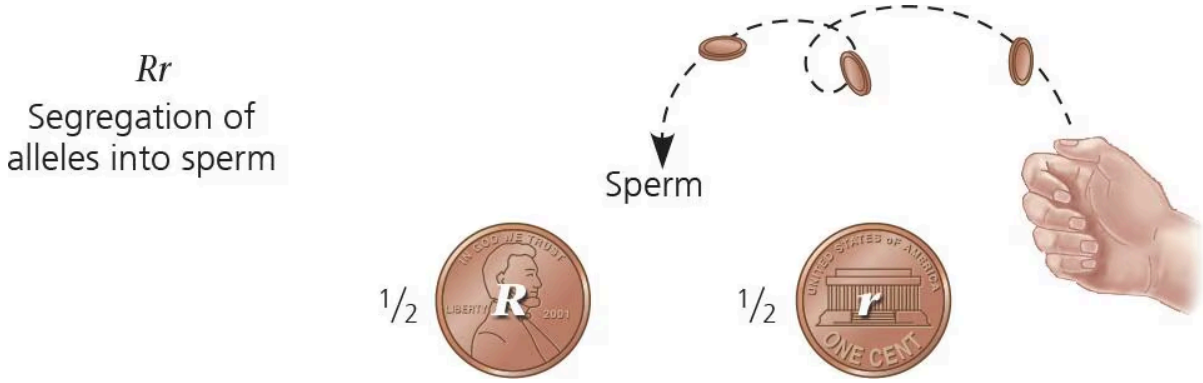
Why was Mendel's choice of pea plants as an experimental organism so fortuitous in developing a model of inheritance? **(BIG IDEA 3)**

Concept 14.1: Mendel used the scientific approach to identify two laws of inheritance

- Gregor Mendel formulated a theory of inheritance based on experiments with garden peas, proposing that parents pass on to their offspring discrete genes that retain their identity through generations. This theory includes two “laws.”
- The ****__law of segregation__ states that genes have alternative forms, or ****__alleles.__ In a diploid organism, the two alleles of a gene segregate (separate) during meiosis and gamete formation; each sperm or egg carries only one allele of each pair. This law explains the 3:1 ratio of F2 ****__phenotypes__ observed when ****__monohybrids__ cross-or self-pollinate. Each organism inherits one allele for each gene from each parent. In ****__heterozygotes__, the two alleles are different: Expression of the ****__dominant allele__ masks the phenotypic effect of the ****__recessive allele__. ****__Homozygotes__ have identical alleles of a given gene and are therefore ****__true-breeding__.
- The ****__law of independent assortment__ states that the pair of alleles for a given gene segregates into gametes independently of the pair of alleles for any other gene. In a cross between ****__dihybrids__ (individuals heterozygous for two genes), the offspring have four phenotypes in a 9:3:3:1 ratio.

When Mendel did crosses of true-breeding purple- and white-flowered pea plants, the white-flowered trait disappeared from the F1 *generation but reappeared in the* F2 *generation. Use genetic terms to explain why that happened.*

Concept 14.2: Probability laws govern Mendelian inheritance



- The ****__multiplication rule__ states that the probability of two or more events occurring together (a statement using *and*) is equal to the product of the individual probabilities of the independent single events. The ****__addition rule__ states that the probability of an event that can occur in two or more independent, mutually exclusive ways (using *or*) is the sum of the individual probabilities.

- The rules of probability can be used to solve complex genetics problems. A dihybrid or other multicharacter cross is equivalent to two or more independent monohybrid crosses occurring simultaneously. In calculating the chances of the various offspring genotypes from such crosses, each character is first considered separately and then the individual probabilities are multiplied.

DRAW IT*Redraw the Punnett square on the right side of**Figure 14.8**as two smaller monohybrid Punnett squares, one for each gene. Below each square, list the fractions of each phenotype produced. Use the rule of multiplication to compute the overall fraction of each of the possible dihybrid phenotypes. What is the phenotypic ratio?*

Concept 14.3: Inheritance patterns are often more complex than predicted by simple Mendelian genetics

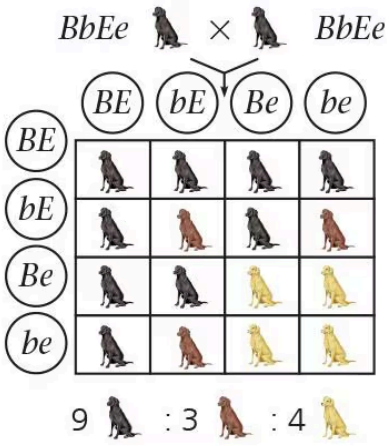
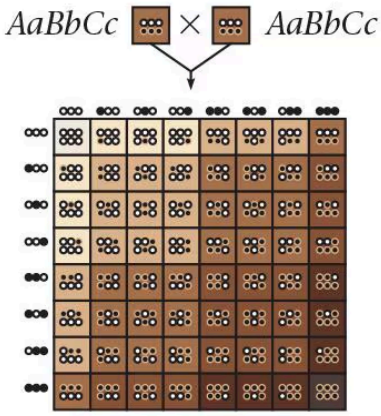
- Extensions of Mendelian genetics for a single gene:

Relationship among alleles of a single gene	Description	Example
Complete dominance of one allele	Heterozygous phenotype same as that of homozygous dominant	PP Pp
Incomplete dominance of either allele	Heterozygous phenotype intermediate between the two homozygous phenotypes	C^RC^R C^RC^W C^WC^W
Codominance	Both phenotypes expressed in heterozygotes	I^AI^B
Multiple alleles	In the population, some genes have more than two alleles	ABO blood group alleles I^A, I^B, i
Pleiotropy	One gene affects multiple phenotypic characters	Sickle-cell disease

Parts are as follows.

1. Complete dominance of one allele: P P both uppercase = purple flowers. Uppercase P lowercase p = purple flowers.
2. Incomplete dominance of either allele: C superscript R, C superscript R = red flowers. C superscript R, C superscript W= pink flowers. C superscript W, C superscript W = white flowers.
3. Codominance: Red blood cell with both A and B (circles and triangles) carbohydrates on the surface in an I superscript A, I superscript B individual.

- Extensions of Mendelian genetics for two or more genes:

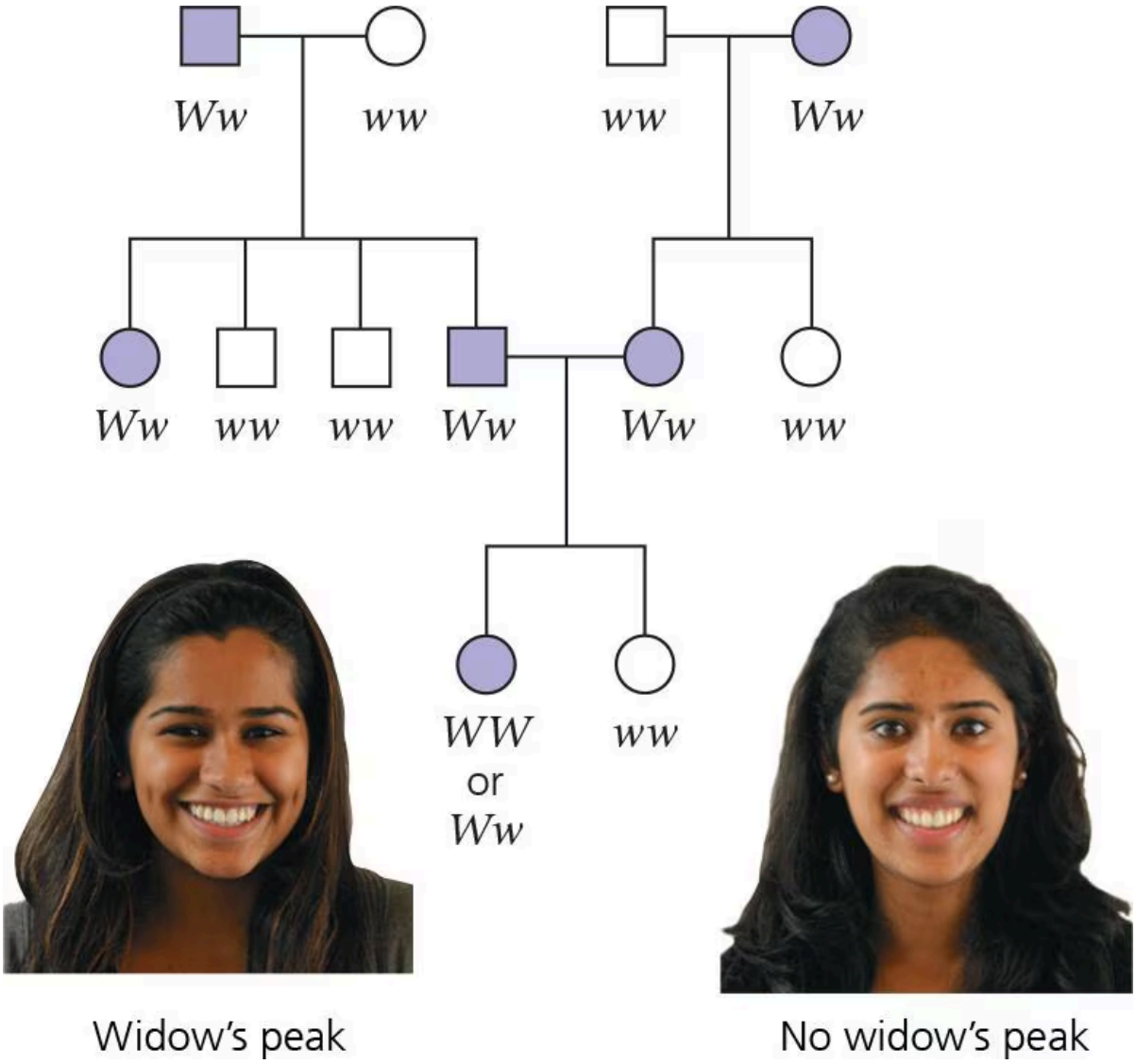
Relationship among two or more genes	Description	Example
Epistasis	The phenotypic expression of one gene affects the expression of another gene	
Polygenic inheritance	A single phenotypic character is affected by two or more genes	

- The expression of a genotype can be affected by environmental influences, resulting in a range of phenotypes. Polygenic characters that are also influenced by the environment are called ******multifactorial** characters.
- An organism's overall phenotype reflects its overall genotype and unique environmental history. Even in more complex inheritance patterns, Mendel's fundamental laws still apply.

Which genetic relationships listed in the first column of the two tables above are demonstrated by the inheritance pattern of the ABO blood group alleles? For each genetic relationship, explain why this inheritance pattern is or is not an example.

Concept 14.4: Many human traits follow Mendelian patterns of inheritance

- Analysis of family ****__pedigrees__ can be used to deduce the possible genotypes of individuals and make predictions about future offspring. Such predictions are statistical probabilities rather than certainties.



Generations are as follows.

First generation (grandparents): Pedigree drawn with symbols as follows from left to right: First grandparents: Shaded Square (uppercase W lowercase w) with line drawn to circle (w w, both lowercase). Space and then second grandparents: square (w w both lowercase) with line drawn to shaded circle

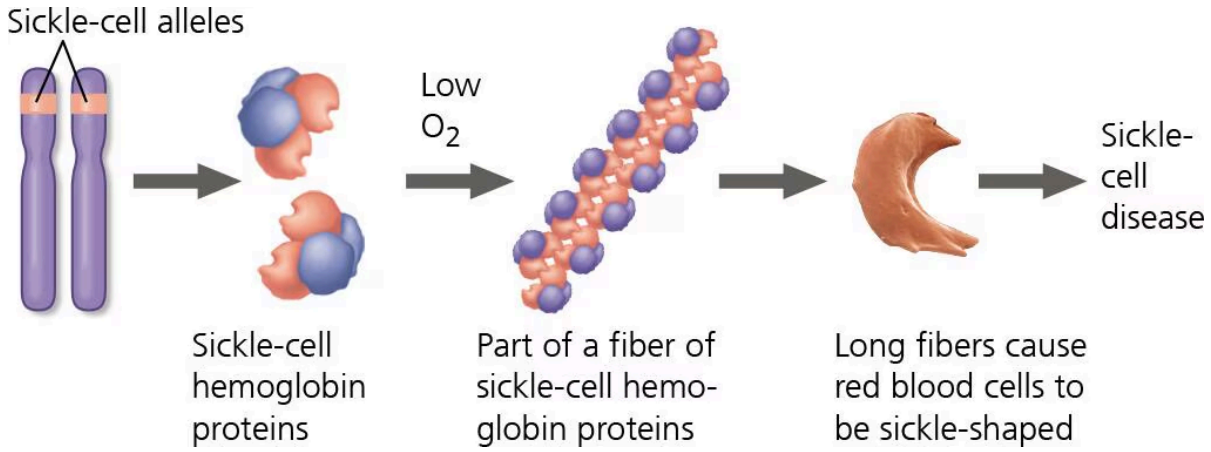
(uppercase W lower case w).

Second generation (parents, uncles and aunts): Children of first grandparents: Shaded circle (uppercase W lowercase w), square (w w both lowercase), square (w w both lowercase), shaded square (uppercase W lowercase w). This individual has a line drawn to the first child of the second grandparents, a shaded circle (uppercase W lowercase w). Next is a circle (w w both lowercase).

Third generation: children of the couple in the second generation: Shaded circle (W W both uppercase or uppercase W lowercase w). This individual is shown next to a photograph of a woman with Widow's peak. Next is a circle (w w both lowercase). This individual is shown next to a photograph of a woman with no widow's peak.

Is a widow's peak a dominant or recessive trait? Tips for pedigree analysis: Notice in the third generation that the second-born daughter lacks a widow's peak, although both of her parents had the trait. Such a pattern indicates that the trait is due to a dominant allele. If it were due to a recessive allele, and both parents had the recessive phenotype (straight hairline), all of their offspring would also have the recessive phenotype.

- Many genetic disorders are inherited as simple recessive traits. Most affected (homozygous recessive) individuals are children of phenotypically normal, heterozygous ****__carriers__.
- The sickle-cell allele has probably persisted for evolutionary reasons: Homozygotes have **sickle-cell disease** ****, but heterozygotes have an advantage because one copy of the sickle-cell allele reduces both the frequency and severity of malaria attacks.



Homozygote with sickle-cell disease: Weakness, anemia, pain and fever, organ damage. Pair of homologous chromosomes, each having a sickle-cell allele, flows to both sickle-cell hemoglobin proteins, flows with low O 2 to part of a fiber of sickle-cell hemoglobin proteins. Long fibers cause red blood cells to be sickle-shaped and causes sickle-cell disease.

- Lethal dominant alleles are eliminated from the population if affected people die before reproducing. Nonlethal dominant alleles and lethal ones that strike relatively late in life can be inherited in a Mendelian way.
- Many human diseases are multifactorial—that is, they have both genetic and environmental components and do not follow simple Mendelian inheritance patterns.
- Using family histories, genetic counselors help couples determine the probability that their children will have genetic disorders. Genetic testing of prospective parents can reveal whether they are carriers of recessive alleles associated with specific disorders. Ultrasounds and blood tests can screen for certain disorders in a fetus. **********Amniocentesis** and **********chorionic villus sampling** can diagnose whether a suspected genetic disorder is present in a fetus. Other genetic tests can be performed after birth.

Both members of a couple know that they are carriers of the cystic fibrosis allele. None of their three children have cystic fibrosis, but any one of them might be a carrier. The couple would like to have a fourth child but are worried that he or she would very likely have the disease, since the first three do not. What would you tell the couple? Would it remove some uncertainty from their prediction if they could find out whether the three children are carriers?

Tips for Genetics Problems

- Write down symbols for the alleles. (These may be given in the problem.) When represented by single letters, the dominant allele is uppercase and the recessive is lowercase.
- Write down the possible genotypes, as determined by the phenotype.
- If the phenotype is that of the dominant trait (for example, purple flowers), then the genotype is either homozygous dominant or heterozygous (PP or Pp in this example).
- If the phenotype is that of the recessive trait, the genotype must be homozygous recessive (for example, pp).
- If the problem says “true-breeding,” the genotype is homozygous.
- Determine what the problem is asking for. If asked to do a cross, write it out in the form [genotype]×[genotype], using the alleles you’ve decided on.
- To figure out the outcome of a cross, set up a Punnett square.
- Put the gametes of one parent at the top and those of the other on the left. To determine the allele(s) in each gamete for a given genotype, set up a systematic way to list all the possibilities. (Remember, each gamete has one allele of each gene.) Note that there are 2n possible types of

gametes, where n is the number of gene loci that are heterozygous. For example, an individual with genotype AaBbCc would produce 23=8 types of gametes. Write the genotypes of the gametes in circles above the columns and to the left of the rows.

- Fill in the Punnett square as if each possible sperm were fertilizing each possible egg, making all of the possible offspring. In a cross of AaBbCc×AaBbCc, for example, the Punnett square would have 8 columns and 8 rows, so there are 64 different offspring; you would know the genotype of each and thus the phenotype. Count genotypes and phenotypes to obtain the genotypic and phenotypic ratios. Because the Punnett square is so large, this method is not the most efficient. See tip 5.
- You can use the rules of probability if a Punnett square would be too big. (For example, see the question at the end of the summary for Concept 14.2 and question 7 below.) You can consider each gene separately (see the section Solving Complex Genetics Problems with the Rules of Probability in Concept 14.2).
- If the problem gives you the phenotypic ratios of offspring but not the genotypes of the parents in a given cross, the phenotypes can help you deduce the parents' unknown genotypes.
- For example, if 12 of the offspring have the recessive phenotype and 12 the dominant, you know that the cross was between a heterozygote and a homozygous recessive.
- If the ratio is 3:1, the cross was between two heterozygotes.
- If two genes are involved and you see a 9:3:3:1 ratio in the offspring, you know that each parent is heterozygous for both genes. Caution: Don't assume that the reported numbers will exactly equal the predicted ratios. For example, if there are 13 offspring with the dominant trait and 11 with the recessive, assume that the ratio is one dominant to one recessive.
- For pedigree problems, use the tips in Figure 14.15 and below to determine what kind of trait is involved.
- If parents without the trait have offspring with the trait, the trait must be recessive and the parents both carriers.
- If the trait is seen in every generation, it is most likely dominant (see the next possibility, though).
- If both parents have the trait, then in order for it to be recessive, all offspring must show the trait.
- To determine the likely genotype of a certain individual in a pedigree, first label the genotypes of all the family members you can. Even if some of the genotypes are incomplete, label what you do know. For example, if an individual has the dominant phenotype, the genotype must be AA or Aa; you can write this as A–. Try different possibilities to see which fits the results. Use the rules of probability to calculate the probability of each possible genotype being the correct one.